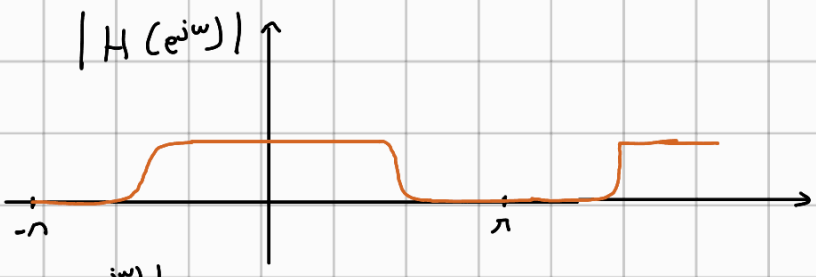
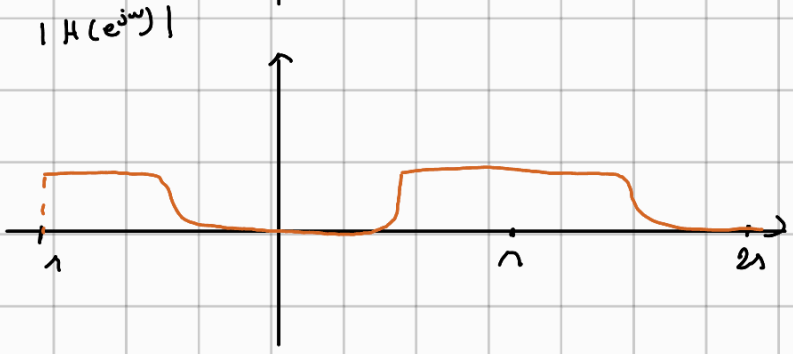
Tip 2

$$H(e^{jw}) = e^{-jw\frac{M}{2}} \underbrace{\sum_{n=0}^{\frac{M-1}{2}} h[n]\left[2\cos\left(wn - w\tfrac{M}{2}\right)\right]}_{A \geq 0}$$

$$\angle H(e^{jw}) = \begin{cases} -w\frac{M}{2} & , A > 0 \\ -w\frac{M}{2} + \pi & , A < 0 \end{cases}$$





$$|H(e^{jw})|_{w=0} = \left|\sum_{n=0}^{\frac{M-1}{2}} 2h[n]\right|$$

$$|H(e^{jw})|_{w=\pi} = \left|\sum_{n=0}^{\frac{M-1}{2}} h[n]\cos\left(\pi n - \tfrac{\pi}{2}M\right)\right| = 0$$

* M tip 2de tektir

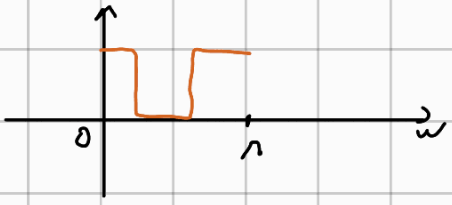
$\cos\left(\frac{\pi}{2}(2n-M)\right)$

çift - tek = tek

1
2
3
4
5

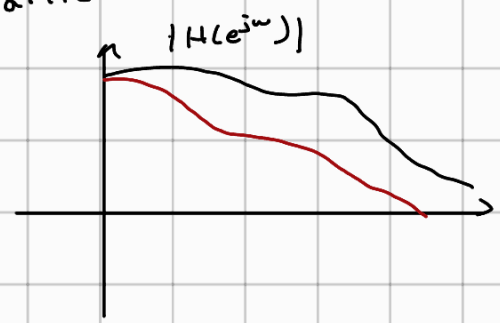
Bant söndüren

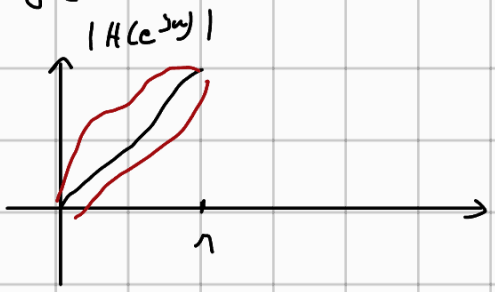
$w = 0$ da ve π de $|H(e^{jw})| > 0$



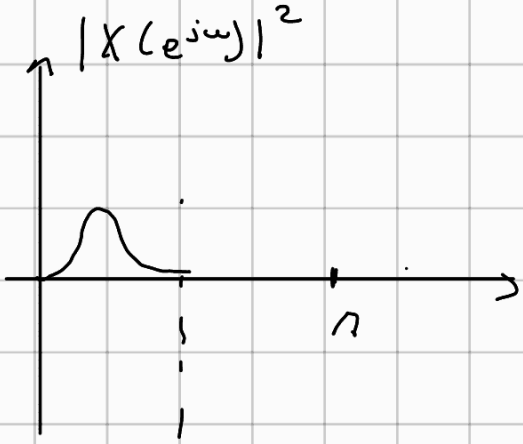
	LP	HP	BP	BS (Band stop)
Type I	✓	✓	✓	✓
Type II	✓	✗	✓	✗
Type III	✗	✗	✓	✗
Type IV	✗	✓	✓	✗

artık $H(e^{j\omega})$ $h[n]$ ün sağlaya yüz









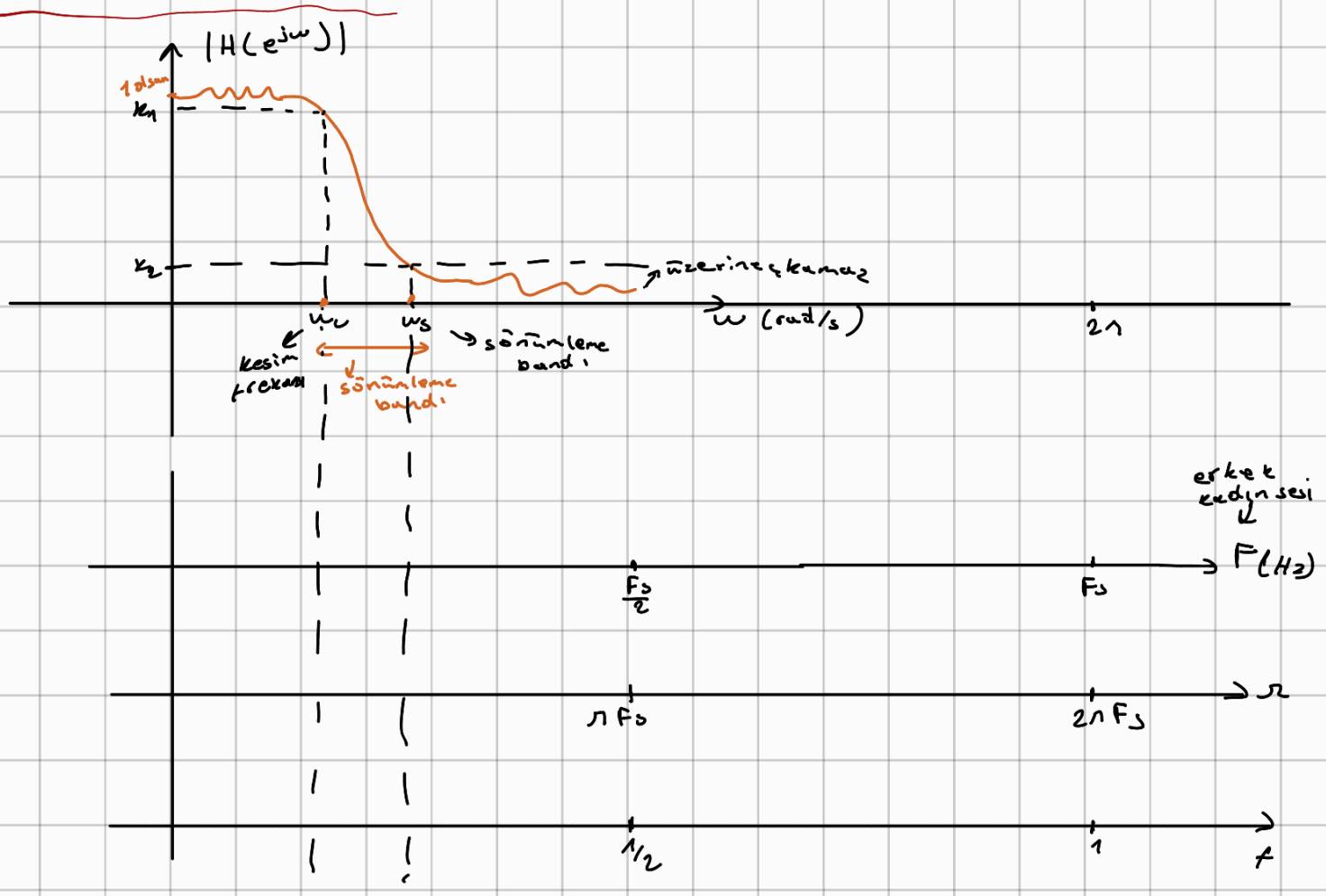




* STFT → iki boyutlu



Filter Specifications



$\Omega = F_s\, w$ $\qquad \Omega = 2\pi F$

$F = F_s \cdot f$ $\qquad w = 2\pi f$

$$w = \frac{F}{F_s}\, 2\pi$$

Tip 1

$h[n] = h[M-n]$ $\qquad M$ even (çift)

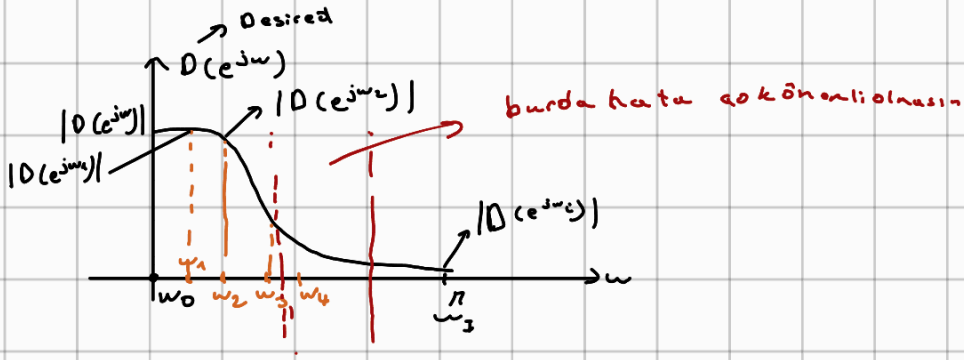
$$H(e^{jw}) = e^{-jw\frac{M}{2}} \left(\sum_{k=0}^{\frac{M}{2}} a_k \cos wk \right)$$

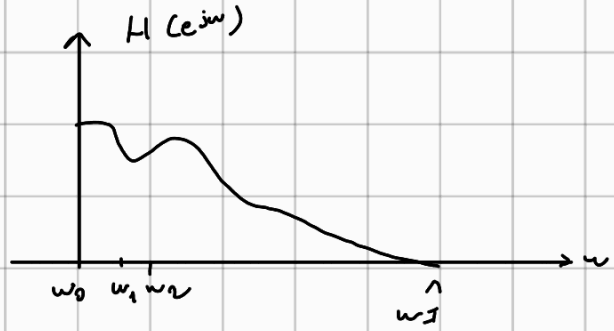
$a_0 = h[M/2]$

$a_k = 2h[M/2 - k]$ $\qquad k = 1, \dots, M/2$

Tip 1 için

$$H(e^{jw}) = e^{-jw\frac{M}{2}} \left(\sum_{m=0}^{M/2} a_m \cos wm \right)$$





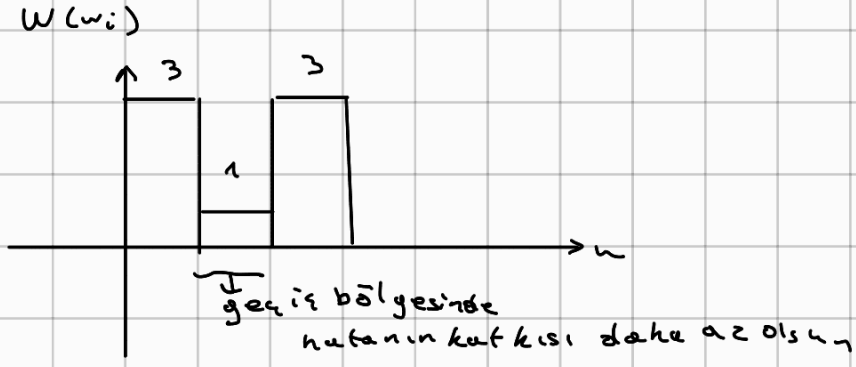
$$\text{Hata} = \left| D(e^{j\omega_k}) - H(e^{j\omega_k}) \right|$$
k noktasında

minimize etmeye gerek

$$\sum_{k=0}^{I} \left(D(e^{j\omega_k}) - H(e^{j\omega_k}) \right)^2$$

kare işlemi yüksek hataları güçlendirir

onlara önem verir



$$\Rightarrow \min \sum_{k=0}^{I} W(\omega_k) \left(D(e^{j\omega_k}) - H(e^{j\omega_k}) \right)^2$$

$$e(\omega_i) = W(\omega_i) \left[H(\omega_i) - D(\omega_i) \right]$$

$$\omega_i = \quad i = 0, \dots I$$

$$E = \sum_{i=0}^{I} |e(\omega_i)|^2$$

$$= \sum_{i=0}^{I} \left[W(\omega_i) \left[H(\omega_i) - D(\omega_i) \right] \right]^2$$

$$= \sum_{i=0}^{I} \left[\sum_{m=0}^{M/2} W(\omega_i) \, e^{-j\omega_i \frac{M}{2}} \cos(\omega_i \cdot m) \, a_m - W(\omega_i) D(\omega_i) \right]^2$$

e_i

Minimize E

$$= e_0^2 + e_1^2 + \dots + e_J^2 = \underbrace{\begin{bmatrix} e_0 & e_1 & \dots & e_J \end{bmatrix}}_{e^T} \underbrace{\begin{bmatrix} e_0 \\ e_1 \\ \vdots \\ e_J \end{bmatrix}}_{e}$$

$$= e^T \cdot e$$

$$e_i = \begin{bmatrix} W(w_i) e^{-jw_i \frac{M}{2}}, & W(w_i) e^{-jw_i \frac{M}{2}} \cos(w_i 1) \end{bmatrix} \underbrace{\begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_J \end{bmatrix}}_{a \text{ vektörü}} - \underbrace{W(w_i) D(w_i)}_{d_i \text{ diyelim}}$$

$$h_i = \begin{bmatrix} 1 \\ \cos(w_i) \\ \cos(2w_i) \\ \cos(3w_i) \\ \vdots \\ \cos\left(\frac{M}{2} w_i\right) \end{bmatrix}$$

$$e_i = W(w_i) e^{-jw_i \frac{M}{2}} h_i^T \cdot a - d_i$$

$$e = \begin{bmatrix} e_0 \\ e_1 \\ \vdots \\ e_J \end{bmatrix} \underbrace{\begin{bmatrix} W(w_0) e^{-jw_0 \frac{M}{2}} h_0^T \\ W(w_1) e^{-jw_1 \frac{M}{2}} h_1^T \\ \vdots \\ W(w_J) e^{-jw_J \frac{M}{2}} h_2^T \end{bmatrix}}_{H \text{ vektörü}} a - \underbrace{\begin{bmatrix} d_0 \\ d_1 \\ d_2 \\ \vdots \\ d_J \end{bmatrix}}_{d}$$

$$e = H a - d$$

minimize $E = e^T \cdot e = (H \cdot a - d)^T (H \cdot a - d) = a^T H^T H a - \underbrace{a^T H^T d}_{a^T \; b} - d^T H a + d^T d$

$$= a^T H^T H a - 2 a^T H^T d + d^T d$$

$a^T b = b^T a$ ✓

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\frac{\partial E}{\partial a} = 0 \qquad \frac{\partial E}{\partial a} = 2 H^T H a - 2 H^T d$$

$$H^T H a = H^T d \rightarrow \boxed{a = (H^T H)^{-1} H^T d}$$

Eğer M'i çok kısıtlı tutarsanız filtrenizin çözünürlüğü düşük